INCH-POUND

MIL-C-82822(OS)
29 December 1989
SUPERSEDING
WS 20890A
15 July 1986

MILITARY SPECIFICATION

CATALYST, BURNRATE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines one type of burnrate catalyst, referred to herein as the catalyst, for use in the manufacture of rocket motor propellant.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-18455 Argon, Technical

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402-0001).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 445 Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity) (DoD adopted)

ASTM D 972 Evaporation Loss of Lubricating Greases and Oils (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

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(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Chemical and physical properties. Chemical and physical properties of the catalyst shall conform to table I.

3.2.1 Infrared spectrum. The infrared (IR) spectrum shall conform qualitatively to the spectrum shown on figure 1.

3.3 Shelf life. The catalyst shall have a shelf life of 12 months when stored at 80 ± 20° Fahrenheit (F) in original sealed containers.

3.3.1 Shelf life extension. The shelf life expiration date of an individual lot of catalyst may be extended for additional 12-month periods provided that the catalyst upon retest successfully meets the viscosity, moisture and insolubility requirements of table I.

3.3.1.1 Repackaging material from opened containers. When a container is opened to obtain material for shelf life extension testing, the material shall be repackaged in the original container or in a suitable, clean, dry, replacement container by subjecting each container to the special packaging, purging, sealing, and security stamping requirements of 5.1.1 before returning the material to storage. The containers shall then be sealed and security stamped.

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TABLE I. Chemical and physical properties.

Property	Units	Limits	
		Minimum	Maximum
Viscosity at 25.0 ± 0.1 degrees Celsius (°C)	Centistokes	---	2200
Iron content	percent	23.3	24.3
Moisture	percent	---	0.10
Acidity	milliequivalent/gram	---	0.025
Insolubility	percent	---	0.1
Evaporation loss	percent	---	4.0

3.4 **Safety.**

3.4.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.4.2 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.4).

3.5 **Workmanship.** Workmanship shall be such that the catalyst is uniform in appearance and free from visible contamination.

4. **QUALITY ASSURANCE PROVISIONS**

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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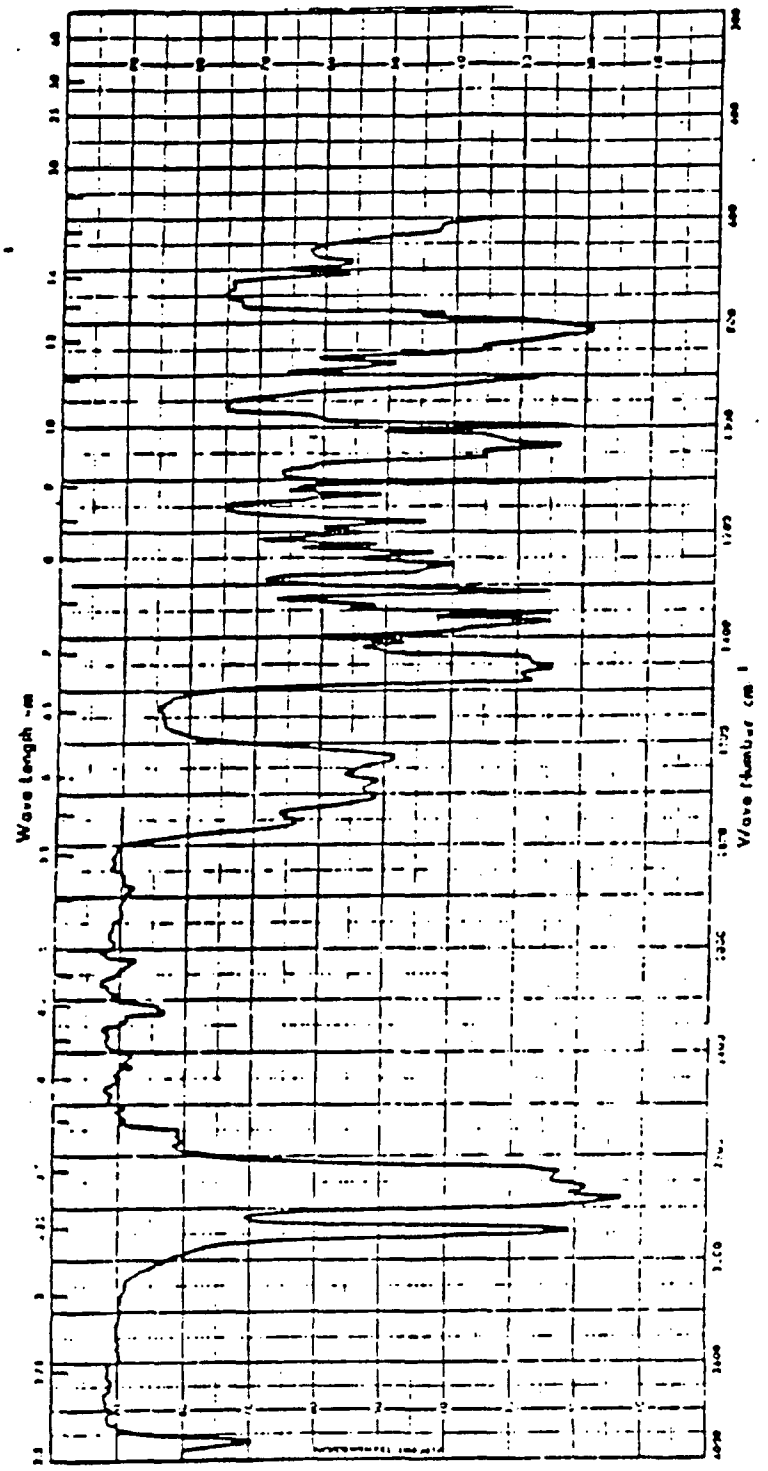


FIGURE 1. Infared spectrum, burnrate catalyst.

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4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the examination and tests specified in 4.4. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** For inspection purposes, a lot shall consist of all the material subject to inspection at one time from one of the following:

a. A single batch (see 6.5.1).

b. A continuous process (see 6.5.2).

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.1.2 **Selection of sample containers.** Sample containers of the catalyst shall be selected at random from each lot in accordance with table III for the inspections specified in 4.5.1 through 4.5.8. Sample material shall be selected as specified in 4.4.1.3.

4.4.1.3 **Sample material.** Sufficient material shall be selected at random from each container to perform all of the inspections specified in 4.5.1 through 4.5.8.

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TABLE II. First article and quality conformance inspections.

Examination and tests	Requirement	Test Method
Visual examination	3.5 & section 5	4.5.1
Viscosity test	3.2	4.5.2
Iron content test	3.2	4.5.3
Moisture test	3.2	4.5.4
Acidity test	3.2	4.5.5
Insolubility test	3.2	4.5.6
Evaporation loss test	3.2	4.5.7
IR spectrum test	3.2.1	4.5.8

TABLE III. Sampling for inspection.

Number of Containers in the lot	Number of sample containers to be selected for inspection
1 - 3	All
4 - 15	3
16 - 25	4
26 - 40	5
41 - 65	7
66 - 110	10
111 - 180	15

NOTE: Mix content of container(s) to be sampled prior to sampling to assure dispersion of any settled solids during sampling.

4.4.1.4 **Noncompliance of samples.** Failure of one or more of the sample containers of material to meet all of the specified requirements shall be cause for rejection of the lot.

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4.4.1.5 **Sample preparation.** The samples selected at random from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites. A single determination of each quality characteristic listed in table I shall be performed on each of the three composite samples independently. When a lot consists of less than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended to form three composite samples. The results of each test shall meet the requirements of table I. Failure of any composite sample to meet any requirement shall be cause for rejection of the lot.

4.5 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined in this specification. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each sample shall be visually examined for conformance to 3.5 and section 5.

4.5.2 **Viscosity test.** Viscosity shall be determined in accordance with ASTM D 445 at 25.0 $\pm$ 0.1°C using a Cannon-Fenske viscometer or equivalent. Record viscosity to the nearest 10 centistokes.

4.5.3 **Iron content test.** Iron content shall be determined in accordance with Method A or Method B as follows:

4.5.3.1 **Method A.**

a. Weigh to the nearest 0.1 milligram (mg) a 0.50 $\pm$ 0.05-gram (g) sample into a flask.

b. Add 10 milliliter (ml) of distilled water and 15 ml concentrated nitric acid. Swirl to complete dissolution.

CAUTION: ADD WATER PRIOR TO ACID OR SAMPLE IGNITION WILL OCCUR.

c. Carefully and constantly swirl the flask while adding 5 ml of concentrated sulfuric acid. Add 1 ml concentrated 70 percent perchloric acid and boiling beads. Heat at low heat and then increase heat to digest at white fumes for approximately 2 to 3 minutes. Add 1 ml concentrated nitric acid and five drops perchloric acid and heat to fumes again. Fume for 3 to 5 minutes.

d. Cool to a safe temperature and add 25 $\pm$ 5 ml of distilled water and reheat until precipitate is completely dissolved.

CAUTION: ADD WATER PRIOR TO ACID OR SAMPLE IGNITION WILL OCCUR.

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e. While the solution is hot, add 10 ml concentrated hydrochloric acid and 5 percent stannous chloride solution dropwise until the solution is colorless and then add one or two drops excess. (Prepare 5 percent stannous chloride by dissolving 25 g of stannous chloride in 50 ml concentrated hydrochloric acid and diluting to 500 ml with distilled water. Add a few granules of tin.)

f. While stirring, add (at one time) 10 ml saturated mercuric chloride solution and let stand for 10 minutes.

g. Add 5 ml phosphoric acid, 100 $\pm$ 20 ml distilled water and three drops of diphenylamine sulfonate indicator and titrate with standard 0.1 N potassium dichromate to a violet end point.

h. Calculation: $\text{Percent Iron} = \frac{V \times N \times 5.585}{W}$

Where: V = Volume of potassium dichromate, ml

N = normality of potassium dichromate

W = weight of sample, g.

$5.585 = \frac{\text{atomic weight Fe}}{10}$

i. Record the iron content to the nearest 0.1 percent.

4.5.3.2 **Method B alternate.**

4.5.3.2.1 **Apparatus.**

a. Titrator, Metrohm E-436 or E-536 or equal with combination platinum electrodes and 10 ml-buret.

b. Beaker, 100-ml.

c. Graduated cylinder, 100-ml.

4.5.3.2.2 **Reagents.**

a. Cupric perchlorate, 0.1 N standardized (AGM-57, procedure supplied by Arapahoe Chemicals, Incorporated).

b. Tetrahydrofuran (THF).

c. Acetonitrile, reagent.

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4.5.3.2.3 **Procedure.**

a. Weigh 0.18 to 0.20 gram of catalyst in a 100-ml beaker.

NOTE: Double the sample size if using a 20-ml buret.

b. Add 25 ml of THF and stir to dissolve. Add 40 ml of acetonitrile and stir.

c. Set up the titrator according to the manufacturer's instructions. Typical conditions for the Metrohm titrator are:

	E-436	E-536
Speed:	3 (dial indication)	10 min/100 percent vol.
Measuring range:	1.5 volt (V)	1.5 V
Mode:	millivolt (mV)	mV

d. Titrate the sample solution against 0.1 N cupric perchlorate using a combination platinum electrode.

NOTE: These solutions are air sensitive, therefore titrate as soon as the sample is in the solution. Do not whip air into the solution while stirring.

e. Locate the equivalent point, determine a blank and calculate as follows:

$$\text{Percent Iron} = \frac{(V-B) \times N \times 5.585}{W}$$

Where: V = volume of cupric perchlorate, ml

B = volume of blank, ml

N = normality of cupric perchlorate

W = weight of sample, g

$$5.585 = \frac{\text{atomic weight Fe}}{10}$$

f. Record iron content to the nearest 0.1 percent.

4.5.4 **Moisture test.** Moisture shall be determined with the following Karl Fischer titration method:

a. Transfer a 10 ± 2-g (weighed to the nearest 0.1 mg) sample to 100 ± 20 ml of neutralized 1:1 pyridine-chloroform carrier solution.

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b. Titrate the moisture using Karl Fischer titration assembly with standardized dilute KF reagent (water equivalence of 1.0 to 1.5 mg of water per ml of KF reagent).

c. Consider determination complete when an end point is reached and remains stable for 30 seconds.

d. Calculate total moisture as follows:

$$\text{Percent moisture} = \frac{VE}{10\,W}$$

Where: V = volume of standardized KF reagent, ml

E = water equivalent of standardized KF reagent, mg/ml

W = weight of sample, g

10 = factor to convert to percentage

e. Record total moisture to nearest 0.01 percent.

4.5.5 **Acidity test.** Acidity shall be determined in accordance with the following:

a. Weigh (to the nearest 0.1 mg) 2.5 $\pm$ 0.5 g of sample in a beaker containing 50 $\pm$ 10 ml methanol.

b. Add 50 $\pm$ 10 ml distilled water, stir to disperse, and titrate potentiometrically using previously standardized 0.1 N NaOH and glass calomel electrode pair.

c. Determine a reagent blank.

d. The end point may be located at the center of the break after plotting EMF versus ml of titration.

e. Calculation:

$$\text{Acidity, (milliequivalents/gram of sample)} = \frac{(A-B)N}{W}$$

Where: A = volume of NaOH titrant used for sample, ml

B = volume of NaOH titrant used for blank, ml

N = normality of NaOH titrant

W = sample weight, g

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f. Record acidity to the nearest 0.001 milliequivalents per gram sample.

4.5.6 **Insolubility test.** Insolubility shall be determined in accordance with the following:

a. Dissolve a 10 ± 2-g sample (weighed to the nearest 0.01 g) in 100 ± 20 ml of reagent grade chloroform.

b. Filter dissolved sample through a tared No. 1 Whatman filter paper (weighed to nearest 0.1 mg). Wash paper and contents with chloroform to remove any soluble sample.

c. Dry filter paper and contents, reweigh to nearest 0.1 mg.

d. Calculation:

$$\text{Percent insoluble matter} = \frac{(A-B)\ 100}{W}$$

Where: A = weight of filter paper and residue, g

B = weight of filter paper, g

W = weight of sample, g

e. Record the insoluble material to the nearest 0.01 percent.

4.5.7 **Evaporation loss test.** Evaporation loss shall be determined in accordance with the following conforming to ASTM D 972:

a. A 10.0 ± 0.5 g sample (weighed to the nearest 0.1 mg) is heated at 210° plus 5, minus 0°F for 22 hours while purging with nitrogen heated to test temperature. Purge at a rate of approximately 2 liters per minute.

b. Calculation: The weight loss is calculated as evaporation loss as follows:

$$\text{Evaporation Loss} = \frac{(A-B)\ \ 100}{A}$$

Where: A = initial sample weight, g

B = sample weight after heating, g

c. Record the evaporation loss to the nearest 0.01 percent.

4.5.8 **Infrared spectrum test.** The infrared spectrum shall be determined using an infrared spectrophotometer in accordance with the following:

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a. Smear a sample of the catalyst onto an infrared window.

b. Rapidly scan the smear between 5 and 15 micrometers (2000 and 650 reciprocal centimeters) without tracing the spectrum. Adjust the smear thickness so that the major absorption peak(s) are between 0 and 30 percent transmittance and the background is between 80 and 100 percent transmittance.

c. Scan the sample from 15 to 2.5 micrometers (650 and 4000 reciprocal centimeters) using instrument settings recommended by the instrument manufacturer for a routine scan.

d. Compare the spectrum obtained with figure 1 and record any deviation.

4.6 **Inspection of packaging.** The catalyst containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified below:

5.1.1 **Level C.** The catalyst shall be packaged to afford adequate protection against loss and contamination and shall be purged with argon conforming to MIL-A-18455 for a minimum of six minutes at a flow rate of 10 standard cubic feet per hour, minimum. The container shall then be air-tight sealed and security stamped before shipment. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special Marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification number, which shall be applied in numerical sequence as the container is filled.

d. Net weight of contents and material net weight.

e. Lot number and date of quality conformance inspection.

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f. Date of manufacture and date of shipment.

g. Applicable toxicity and hazard warnings in accordance with the requirements of the applicable Federal Statutes and Regulations.

h. Shelf life and inspection/test date for extension. In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR ADDITIONAL 12 MONTH PERIODS PROVIDED THE VISCOSITY, MOISTURE AND INSOLUBILITY, UPON RETEST, CONFORM TO THE REQUIREMENTS SPECIFIED IN MIL-C-82822."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The material specified herein is intended for use as a burnrate catalyst in the propellant for Standard Missile Mark 104 Dual Thrust Rocket Motors.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1, 2.1.1 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. Material Safety Data Sheet (see 3.4.2 and 6.4).

e. Packaging required if different from 5.1.1.

f. Special marking, if other than as specified in 5.2.2.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the catalyst sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the

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right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Material Safety Data Sheet (MSDS). MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MSDSs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Definitions.

6.5.1 Batch. A batch is that amount of burnrate catalyst made as one unit in the manufacturing process.

6.5.2 Continuous process. A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the material is accumulated for a lot.

6.6 Possible material. The following material has been found to meet the requirements of this specification. This is only given for information and is not restrictive.

Catocene - manufactured by Arapahoe Chemicals

6.7 Subject term (key word) listing.

Propellant
Rocket Motor, Dual Thrust, Mark 104
Rocket Motor Propellant
Standard Missile

Preparing activity:
NAVY-OS
(Project 6810-N057)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-82822	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
CATALYST BURNRATE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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